**NAME** : A.K. LIKITH CHARAN
**REG NO** : 192412305
**COURSE NAME** : ANALOG AND DIGITAL COMMUNICATION (ECA0802)

**EXPERIMENT NO: 3A**

**GENERATION OF PHASE MODULATED (PM) SIGNALS AND CALCULATION OF PHASE DEVIATION USING MATLAB SIMULATION**

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**OBJECTIVES**

i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.

ii) To observe and analyze the time-domain characteristics of the message signal, carrier signal, and phase modulated signal through MATLAB simulation.

iii) To calculate the phase deviation of the PM signal using the phase sensitivity constant and message signal amplitude.

iv) To determine the modulation index of the PM signal and examine its relationship with phase deviation.

v) To investigate the effect of varying message signal amplitude on the phase deviation and modulation characteristics of the PM signal.

vi) To compare PM waveforms obtained for different phase deviation values and analyze their modulation behavior using MATLAB.

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**PROGRAM (MATLAB CODE)**

```
clc;

clear;

close all;

% Parameters

Am = 1;      % Message Amplitude (V)

fm = 100;     % Message Frequency (Hz)

Ac = 5;      % Carrier Amplitude (V)

fc = 1000;    % Carrier Frequency (Hz)

fs = 20*fc;   % Sampling Frequency (Hz)

kp = pi/2;    % Phase Sensitivity (rad/V)

% Time Vector

t = 0:1/fs:4/fm;
```

```
% Message Signal
m = Am*sin(2*pi*fm*t);
% Carrier Signal
c = Ac*cos(2*pi*fc*t);
% Phase Modulated Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
% Phase Deviation
phase_dev = kp*Am;
% PM Modulation Index
mp = phase_dev;
% Display Results
fprintf('Message Amplitude = %.2f V\n',Am);
fprintf('Message Frequency = %.0f Hz\n',fm);
fprintf('Carrier Amplitude = %.2f V\n',Ac);
fprintf('Carrier Frequency = %.0f Hz\n',fc);
fprintf('Phase Sensitivity = %.2f rad/V\n',kp);
fprintf('Phase Deviation = %.2f radians\n',phase_dev);
fprintf('PM Modulation Index = %.2f\n',mp);
% Plotting
figure('Name','Phase Modulation and Phase Deviation','NumberTitle','off');
subplot(2,2,1);
plot(t,m,'LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,2);
plot(t,c,'LineWidth',1.5);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

```
grid on;
subplot(2,2,3);
plot(t,pm,'LineWidth',1.5);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,4);
bar(phase_dev);
title('Phase Deviation');
ylabel('Radians');
set(gca,'XTickLabel',{'PM Signal'});
grid on;
```

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**PROCEDURE**

1. Open MATLAB and create a new script file.
2. Initialize the MATLAB workspace using:
   **clc; clear; close all;**
3. Define the message signal amplitude (A_m), message frequency (f_m), carrier amplitude (A_c), carrier frequency (f_c), sampling frequency (f_s), and phase sensitivity constant (k_p).
4. Generate the time vector using the sampling frequency.
5. Generate the message signal using a sinusoidal waveform.
6. Generate the carrier signal using a sinusoidal waveform.
7. Generate the Phase Modulated (PM) signal using:
   ( s(t) = A_c \cos(2\pi f_c t + k_p m(t)) )
8. Plot the message, carrier, and PM signals.
9. Observe phase variations with respect to the message signal.
10. Calculate the phase deviation using:
    ( \Delta \theta = k_p \cdot A_m )
11. Determine the PM modulation index.
12. Display results using **fprintf()**.
13. Repeat for different amplitudes and observe changes.

14. Compare PM waveforms for different deviations.
15. Record observations and verify relationships.
16. Analyze and validate PM characteristics.

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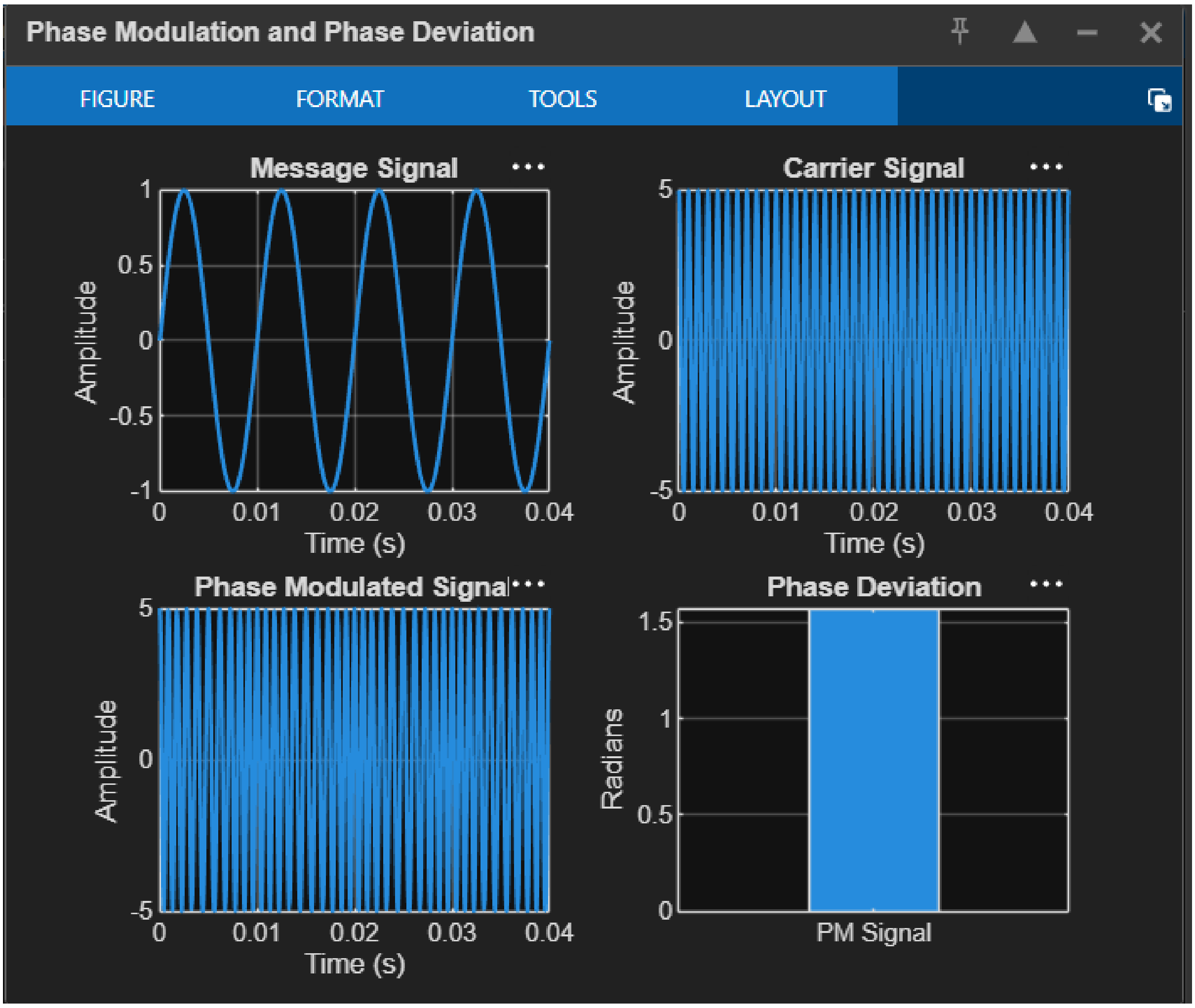
**OUTPUT**




**RESULT**

The Phase Modulated (PM) signal was successfully generated and analyzed using MATLAB. The phase deviation was calculated from the phase sensitivity constant and message amplitude.

i) Message Frequency = 100 Hz
ii) Carrier Frequency = 1000 Hz
iii) Phase Sensitivity = 1.57 rad/V
iv) Phase Deviation = 1.57 radians
v) PM Modulation Index = 1.57
vi) The PM waveform exhibited phase variations proportional to the message signal amplitude, verifying the characteristics of Phase Modulation.